Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/24/2024
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Project Title:	Pool Boiling Heat Transfer Mechanisms in Low Gravity: Numerical Experiments of MABE and NPBX Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balaras, Ilias Ph.D. (George Washington University)		
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Task Description:	The key challenge in developing a proper understanding of nucleate boiling in thermal systems for space environments is the accurate characterization of bubble dynamics and associated heat transfer mechanisms over a wide range of wall superheat and liquid subcooling conditions. To address this challenge, we are proposing well-characterized numerical experiments using high fidelity Direct Numerical Simulation for investigating the dynamics of nucleate boiling across a range of gravity conditions. Our numerical experiments will recreate the conditions of the Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) that were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. The focus of NPBX was the investigation of how bubbles grow and merge in microgravity. The main finding was that heat fluxes for steady state nucleate boiling were found to be substantially lower than what was expected on the basis of low gravity data obtained on Earth. The numerical solver that we will use to simulate MABE conditions has been recently validated by comparing the heat flux from our fully 3-D simulations to MABE data for a representative sample of wall superheat and liquid subcooling conditions. Details about our numerical methods and validation case studies have been recently published (Akash Dhruv, Elias Balaras, Amir Riaz, and Jungho Kim. A formulation for high-fidelity simulations of pool boiling in low gravity. International Journal of Multiphase Flow, 120:103099, 2019). These case studies demonstrate that our solver is able to resolve fully the complex interactions of liquid and vapor phases involving phase transition, interphase mass flux, and contact line dynamics. Using the data from the simulations, we have been able to connect the dynamics of bubble nucleation and growth to the corresponding heat flux measurements in MABE data. Such detailed characterizations of the flow field have already started to shape and improve our understanding of the underlying bubble dynamics. In the proposed research, we aim to systematically probe the MABE database now at a deeper level to extract fundamental information on: (i) how is heat flux partitioned into contributions arising from nucleate boiling and natural convection? (ii) what is the role of contact angle in affecting the force of wall adhesion? and (iii) how much heat transfer occurs through microlayers in low gravity settings? These questions will be addressed to qualitatively different regimes of bubble dynamics under various conditions of gravity, wall superheat, and liquid subcooling. The answers to these questions, investigated on the basis of a validated solver using MABE data, would extend the general understanding of pool boiling in low gravity settings. The use of the MABE database is instrumental to this endeavor. It will be used to provide the reference dataset that connects heat flux, and in some cases even visualization of nucleate boiling, to different levels of gravity, wall super heat, and subcooling. This will allow a unique opportunity for a quantitative assessment of potential improvements in the design of thermal systems in space applications. The proposed research is expected to complement NASA's ongoing efforts, in addition to advancing the science of heat flux partitioning, contact line dynamics, and the role of microlayers for nucleate boiling in low gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Electronic devices require effective and efficient means of thermal management to achieve performance targets. The use of latent heat removal via the process of phase change, such as nucleate and pool boiling, is an extremely effective method of thermal management. This method relies primarily on the dominance of buoyancy over capillary forces to maintain high heat flux at the heated surface. The proposed research will study the influence of major parameters on the behavior of boiling heat transfer. The resulting data and analysis will help improve the thermal management of electronic devices, enabling the achievement of higher performance goals.
Task Progress:	In this reporting period, numerical simulations were carried out using experimental conditions of the Microheater Array Boiling Experiment (MABE) dataset. Simulations were performed on the NASA Advanced Supercomputing (NAS) system. Each simulation used around one thousand cores and ran for approximately 200 hours. One conference paper was published. One full length manuscript was submitted to the International Journal of Heat and Mass Transfer. It has cleared the first round of review with a request for “minor revision". The revision has been submitted. A second journal manuscript is under preparation and will be submitted shortly. Two graduate students presented their work at the TFAWS 2024 (the Thermal & Fluids Analysis Workshop). One student received the best poster award. One student is scheduled to attend the 2024 the annual conference of the American Physical Society – Division of Fluid Mechanics. Task 1. Study of the effect of dimensionless parameters on subcooled nucleate pool boiling. Numerical experiments were caried out using FC-72 as a working fluid with the objective of understanding the effect on subcooling. In the first set of experiments, the bulk liquid temperature was varied while maintaining all other parameters constant. For the second set, the Stefan number was held constant and the effect of changing the wall superheat and bulk liquid temperature was studied. The results from these two sets of data were analyzed to understand how subcooling affects the boiling curve, the heat flux, and bubble dynamics. The primary objective was to test the assumption in the literature that the effect of subcooling was not significant. It was found that the heat flux decreases with an increase in subcooling until it reaches a minimum value beyond which it increases again. Bubble dynamics were studied and coupled to the effect of subcooling on heat flux. The decrease of subcooling before the minima was found to be due to turbulence induced by the departing bubbles, while the subsequent increase of heat flux is due to bubble coalescence and shrinking at the wall. Task 2. Study of the effect of heterogenous wettability surfaces on the thermal performance of nucleate boiling. Numerical experiments of pool boiling were performed to help identify the key physical mechanisms that are responsible for enhancing heat transfer on biphilic surfaces. A biphilic parameter, defined as the ratio of hydrophobic surface area to total surface area (A*=" Hydrophobic area / Total area") was varied and its effect on the rate of heat transfer was investigated. Three cases of A^*= 0.1, 0.25 and 0.75 were analyzed. The highest cumulative heat transfer rate was observed for A^*=0.25. Key observations from the study are that (i) due to the difference in contact angles on hydrophobic and hydrophilic islands, bubbles are drawn towards the hydrophobic islands where they become confined, (ii) Non-classical, poly-anchored vapor structures attached to multiple heater locations are formed that trap thin evaporating liquid layers, and (iii) Instantaneous heat flux partitioning on the heater shows that the poly-anchored vapor structures are responsible for the high heat transfer on biphilic surfaces. Heat partitioning analysis was used to quantify the relative contribution of different mechanisms of heat transfer. Heat

transfer in poly-anchored structures (PLS) and around the contact line are high heat flux mechanisms. They are combined together in the calegory of interface mechanism. The Q-criterion and turbulent fluctuations suggest that the interfacial mechanism is not accompanied by high-energy flow, which is correlated more closely with the mechanisms of quenching and liquid convection. The interface mechanism is responsible for the high heat transfer associated with the intermediate area ratio. While the quenching mechanism is dominant in the large area ratio case, it does not occur over an area large enough to be significant. Similarly, while the interface mechanism occurs in the low area ratio case, the dominant mechanism there is liquid convection. The analysis of the bubble size distribution indicales that the capillary length scale divides the bubble population into two groups that are individually affected by the area ratio.

In this study, the distance between hydrophobic spots is varied as a sole parameter. It is however clear from the analysis presented here that the formation of PLS related hotspots depends on factors such as surface tension, liquid subcooling, wall superheat, number of nucleation sites, and the bubble waiting time. These factors are expected to influence the selection of the optimal conditions related to the rate of bubble growth and the size of the biphilic array. The current study represents a first step towards the fuller understanding of how these other factors combine to influence PLS and heat transfer characteristics on biphilic surfaces.

Bibliography Type: Description: (Last Updated: 04/23/2026)

Abstracts for Journals and Proceedings

Youssoufi S, Riaz A, Balaras E. "Heat transfer and bubble dynamics behaviors in subcooled pool boiling at Earth gravity and ISS gravity using high-fidelity simulations." TFAWS 2024 (the Thermal & Fluids Analysis Workshop), Cleveland, Ohio, August 26-30, 2024.
Abstracts. TFAWS 2024 (the Thermal & Fluids Analysis Workshop), Cleveland, Ohio, August 26-30, 2024. , Aug-2024